

Long (PP) (2016-2017)

Q. Write note on FP Growth and explain it with the help of example.

Q. Write note on naïve bayes classifiers.

Q. Discuss few techniques of data preprocessing.

<u>k-means</u>	<u>k-medoids</u>
Centroids:	Actual data
o Calculated as the mean of data points	point chosen as cluster centers
Sensitive to noise:	
o Sensitive to outliers & noise.	Robust to outliers & noise
Distance metric:	
o Typically uses Euclidean distance	use any distance metric
Computational cost:	
o faster / more	slower / less

Cluster Representation:

Centroids use
Abstract point

categorical data handling:

Requires numerical
data.

cluster shape:

spherical &
isotropic

medoids use
actual data
points.

can handle
categorical data

can handle non
- spherical &
non-isotropic

Q. Use agglomerative clustering
to cluster the following
data.

$(2,4)$, $(5,6)$, $(6,5)$, $(5,8)$,
 $(6,6)$, $(7,7)$, $(2,8)$

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Difference b/w
and clustering?

classification

classification

clustering

Objective:

- Predict the class or category of a new instance.

- Organize data into groups based on similarity.

Supervision:

- Supervised learning

- Unsupervised learning

Training data:

- Requires labeled data

- Does not require labeled data.

o/p:

- Assign instances to predefined classes

- Group similar instance together

Evaluation metrics:

- Accuracy, precision, recall, F1 score.

- Silhouette score

Decision boundary:

- well defined b/w DB class

- No predefined D.B.

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MTWTFSS

Ex:

- Spam email detection, sentiment analysis.

Algorithms:

- Decision tree
- Logistic regression
- ANNs

- Customer segmentation

- K-means
- Hierarchical
- DBSCAN

Q. Discuss significance and applications of data mining in real life.

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Q. Write note on statistical methods.

1) Parametric:

Parametric methods use a fixed number of parameters to build the model.

These methods typically assume that the data follows a known probability distribution such as ND and estimate the parameters of this distribution.

Assumptions:

Require several assumptions about the data:

- Normality
- Homogeneity of variance
- Independence

Test / methods:

- T-test - difference b/w means of two independent groups
- ANOVA - Difference b/w means of multiple groups
- F-Test - compares variance

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- Linear regression
- Logistic regression
- Naive Bayes

2) Non-Parametric:

Do not rely on specific assumptions about the underlying distribution of the population being studied. are distribution free methods. They make no assumption about the shape of the distribution.

Assumption:

- Independence
- Random sampling
- Homogeneity of measurement

Test:

- Spearman's rank correlation

Model:

- KNNs
- Decision tree
- SVM
- ANN

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Q. Difference between supervised and unsupervised learning?

	SML	USML
i/p data:	Algorithms are trained using labeled data.	Use unlabeled data for training.
complexity:	Simplex method	Computationally complex
Accuracy:	High accurate	Less accurate
No of classes:	No of classes is known	No of classes is unknown
Deployment:	uses offline analysis	uses real-time analysis
model:	We can test model.	can not test model
Algorithms:	Classification & Regression	Clustering & Association

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Write pros of supervised ML?

- Not suitable for handling complex tasks.
- Not predict the correct o/p if the data is different from training.
- Require lot of computation times for training.
- Require knowledge of class.

Write cons of supervised ML?

- Explicit feedback
- Predictive Accuracy

Write pros of unsupervised ML?

- Not require prior knowledge.
- Suitable for complex task.

Write cons of unsupervised ML?

- Difficult
- Less accurate
- Not give exact o/p

Q. What is data visualization?

o Data visualization is the graphical representation of information and data.

o DV Translates complex data sets into visual formats that are easier for the human brain to comprehend.

o Tools:

- Charts
- Graphs
- Maps
- Dashboards

o Main goal: is to interpret easily and to make data more accessible.

Q. List types of data for visualization?

1. Numerical → continuous
→ discrete

2. Categorical → binary
→ nominal
→ ordinal